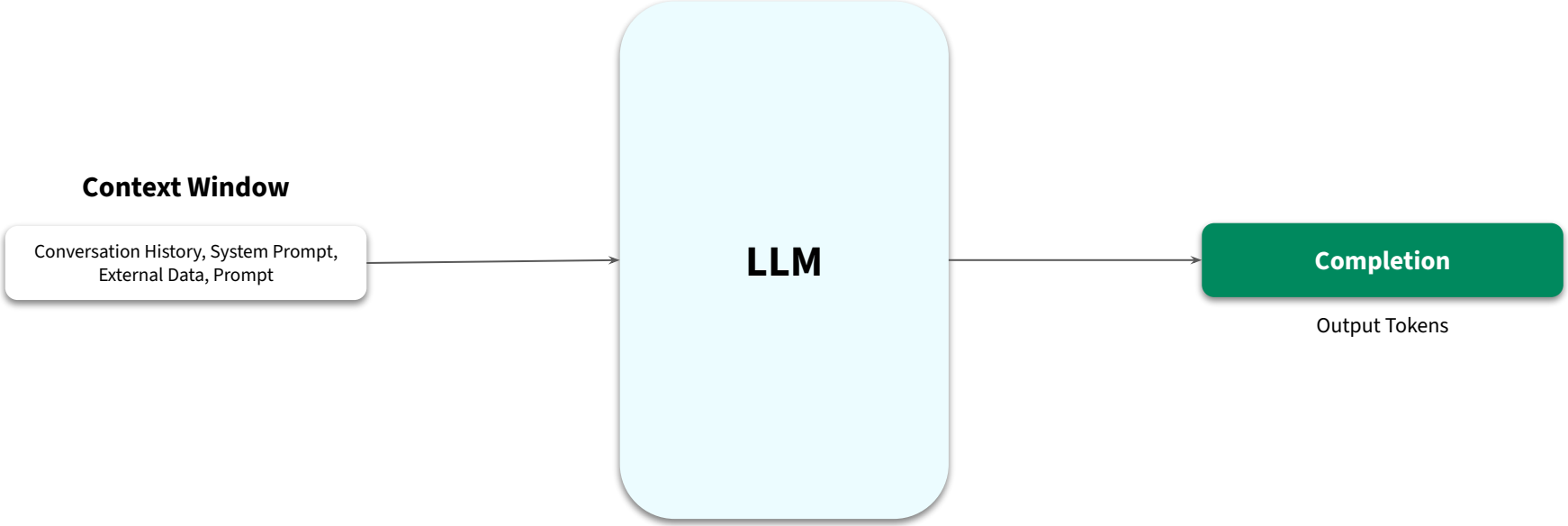
GeeksforGeeks

Why Context Matters?

Retrieval-Augmented Generation

Context Window: The Model's Memory



Good Context = Good Results

Does the new 'Aether-5' drone have 4K video recording?

User

The Aether drone series is known for its excellent camera quality. Most modern drones in its class support 4K video, so it is highly likely that the Aether-5 does as well.

Chatbot

Does the new 'Aether-5' drone have 4K video recording?

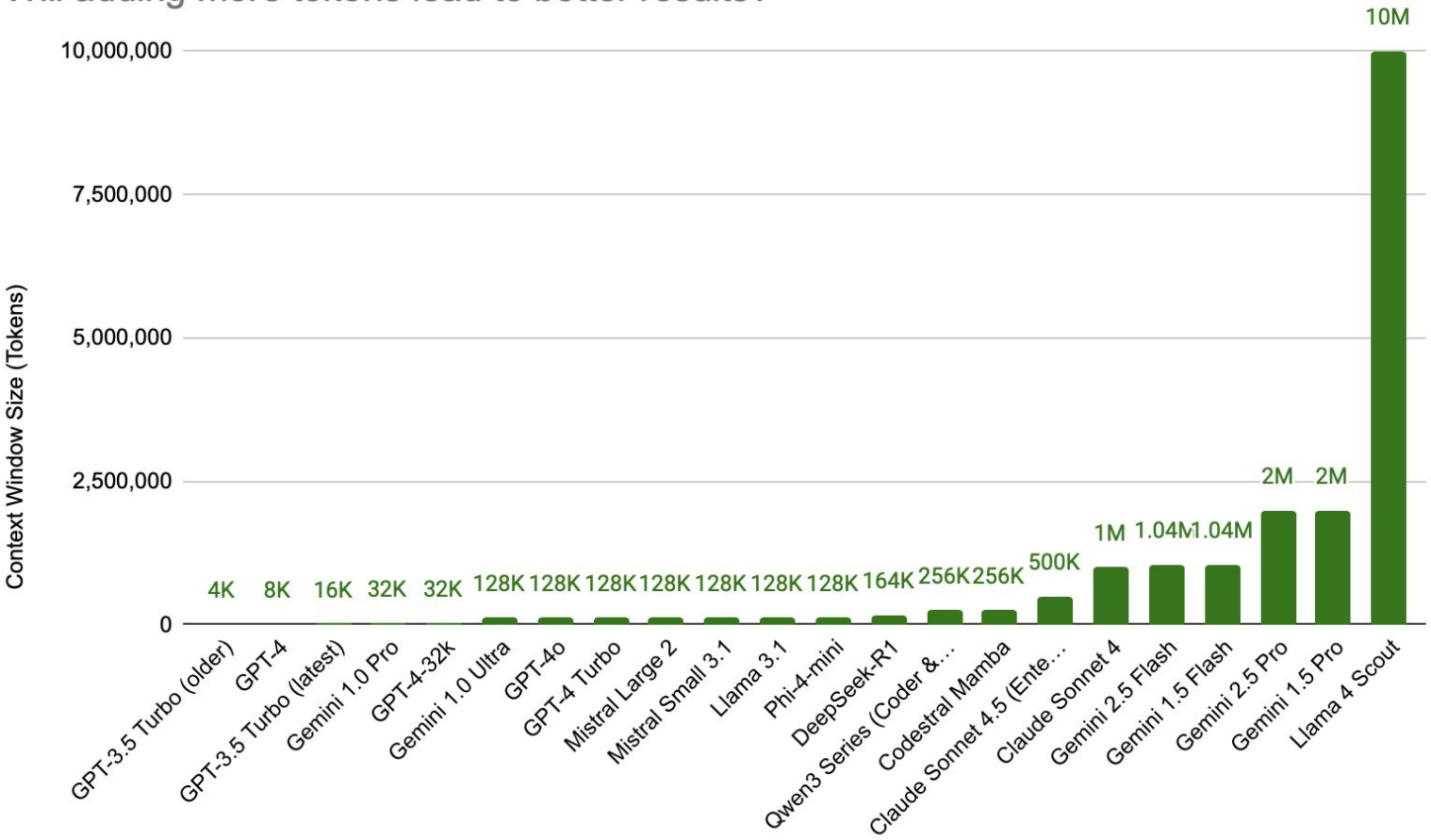
User

Context: The Aether-5 drone, launched in September 2025, features a new 1-inch sensor. This enables video capture in 6K at 30fps and 4K at up to 120fps, a significant upgrade from the Aether-4's 1080p capability

Yes, the Aether-5 drone supports 4K video recording at up to 120fps. It can also record in 6K at 30fps.

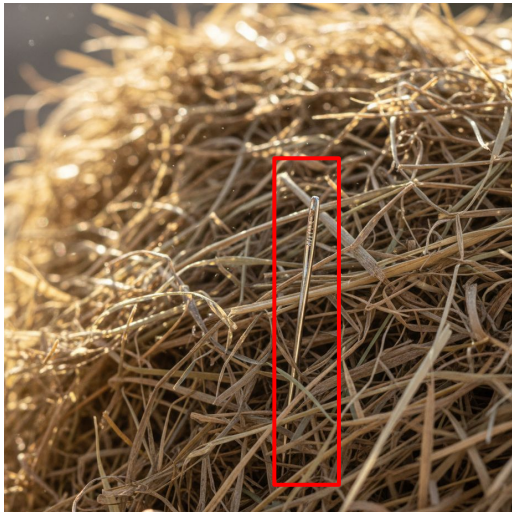
Chatbot

Will adding more tokens lead to better results?



No.

Needle in the Haystack



Good, Small Context = Good Results

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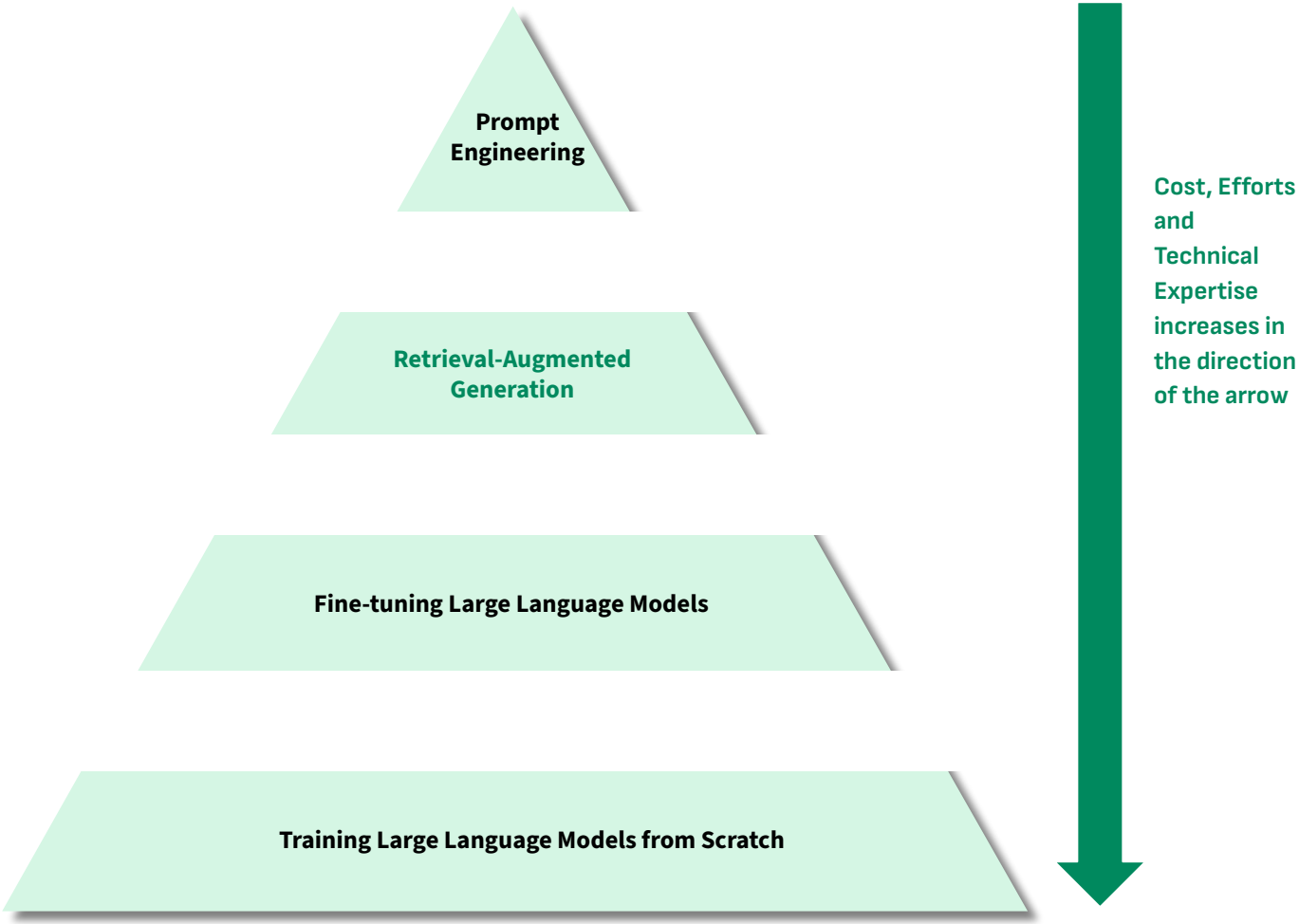
When to use RAG

Retrieval-Augmented Generation

Business Factors

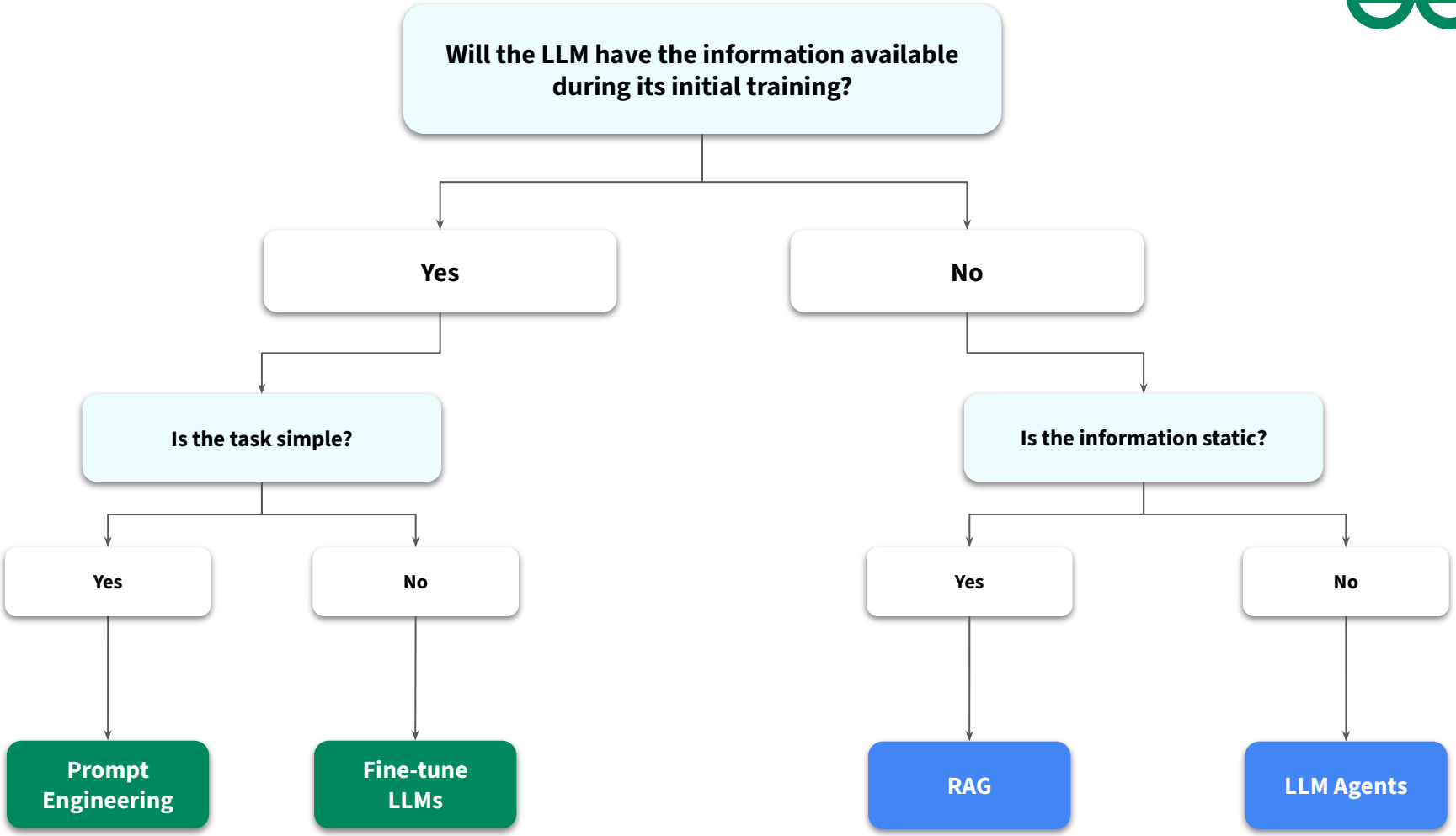
Business Factors

- Cost
- Efforts
- Technical Expertise


Prompt Engineering
Retrieval-Augmented Generation
Fine-tuning Large Language Models
Training Large Language Models from Scratch
Cost, Efforts and Technical Expertise increases in the direction of the arrow

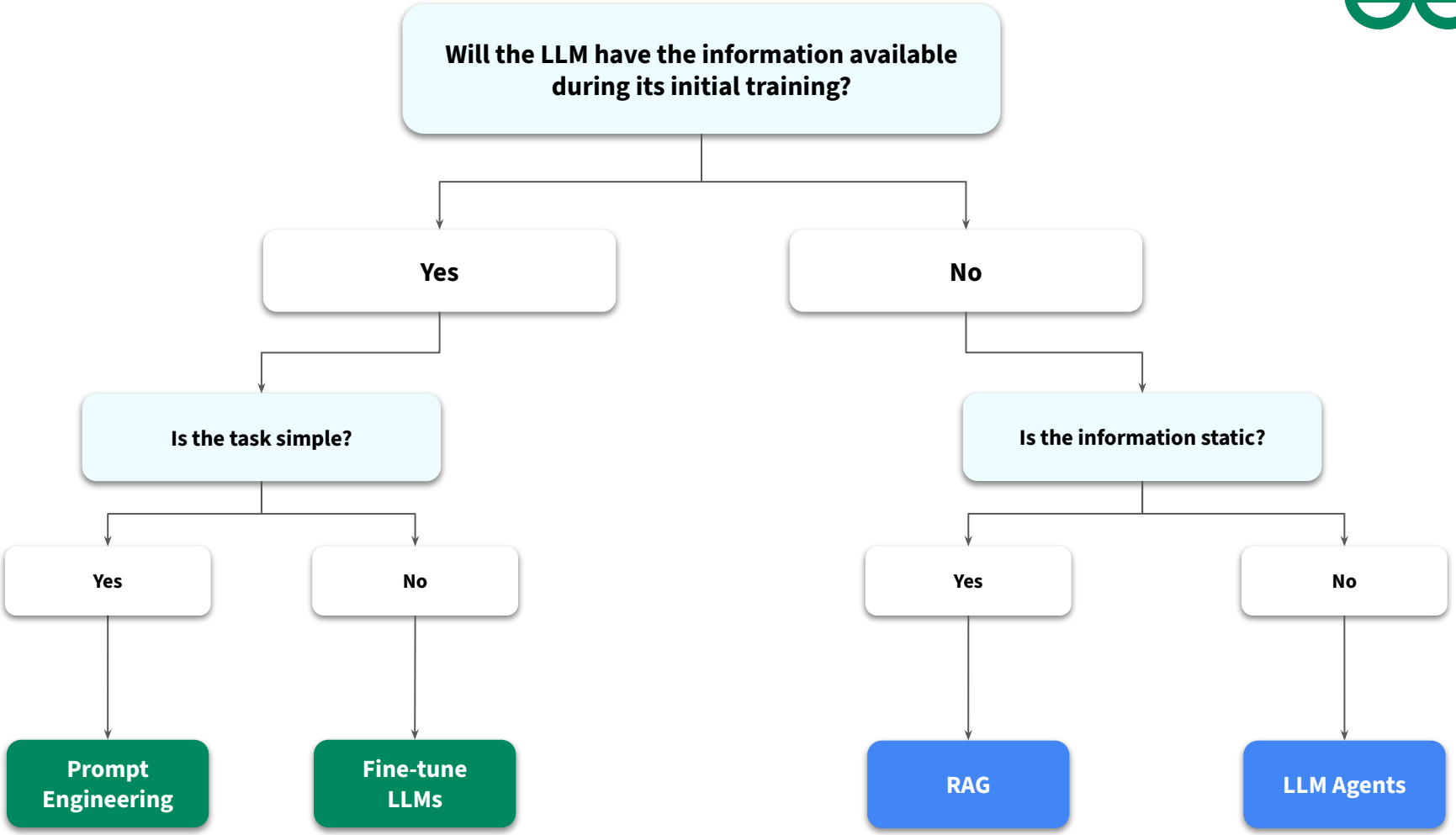
Technical Perspective

Information Required for the Task


Will the LLM have the information available during its initial training?
Yes
No
Is the task simple?
Is the information static?
Yes
No
Yes
No
Prompt Engineering
Fine-tune LLMs
RAG
LLM Agents

Scenario One

You want to use LLMs to generate medical diagnosis using medical records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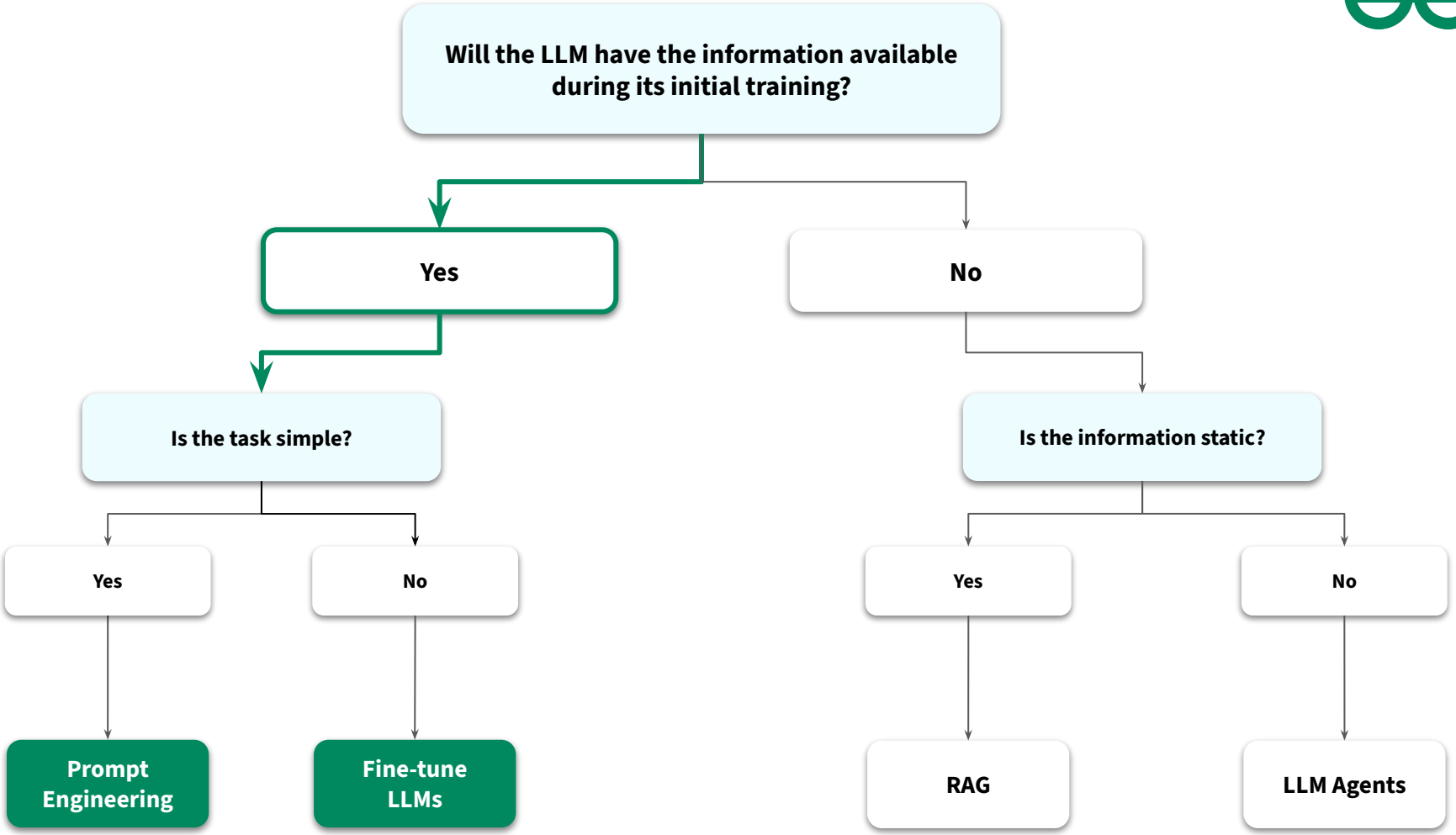
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RAG
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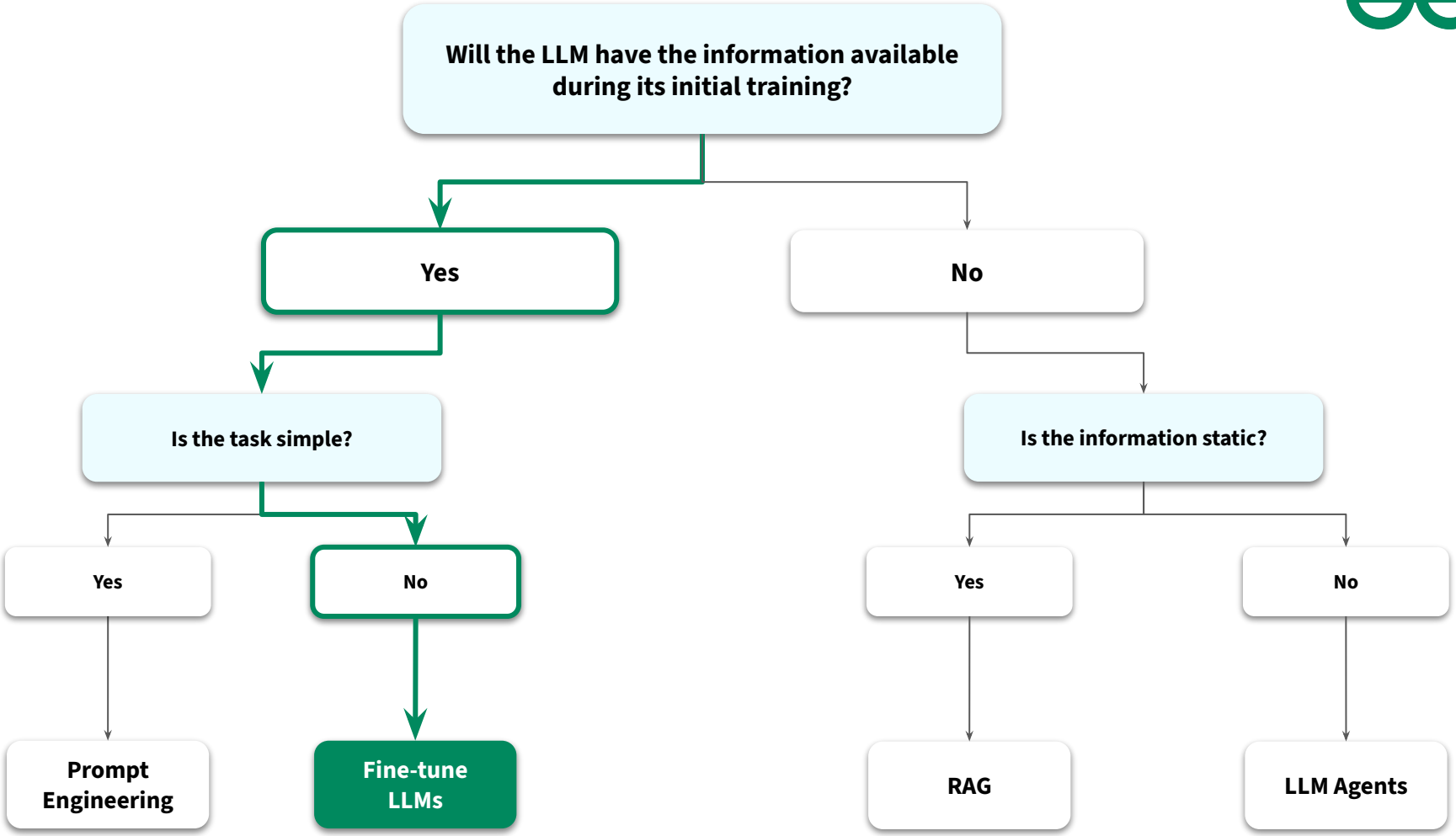
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Yes
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Is the task simple?
Is the information static?
Yes
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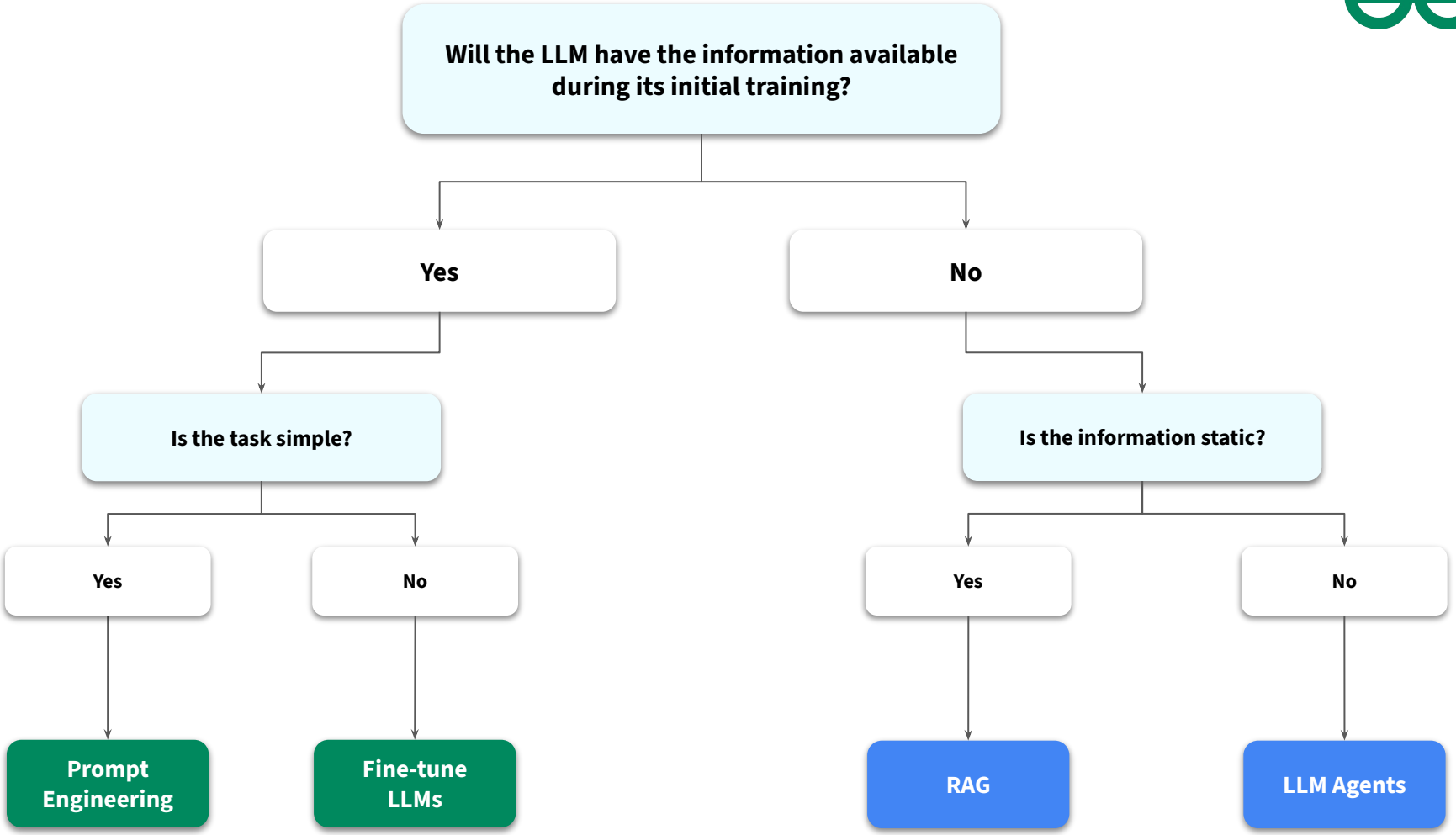
Yes
No
Prompt Engineering
Fine-tune LLMs
RAG
LLM Agents

Scenario Two

You want to develop a chatbot that helps your team access all company policy documents and answers any related questions.


Will the LLM have the information available during its initial training?
Yes
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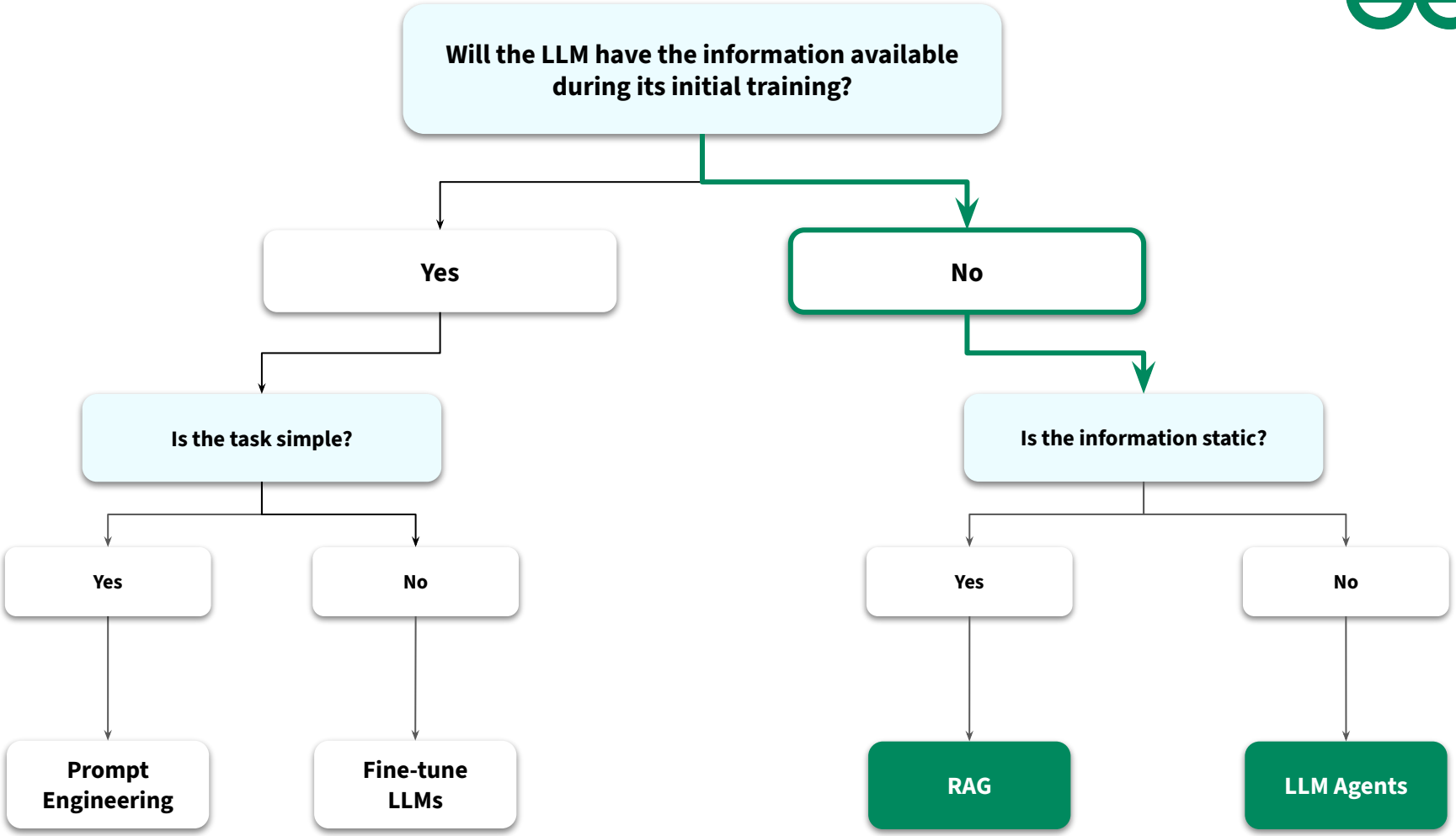
Is the task simple?
Is the information static?
Yes
No
Yes
No
Prompt Engineering
Fine-tune LLMs
RAG
LLM Agents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Is the task simple?
Is the information static?
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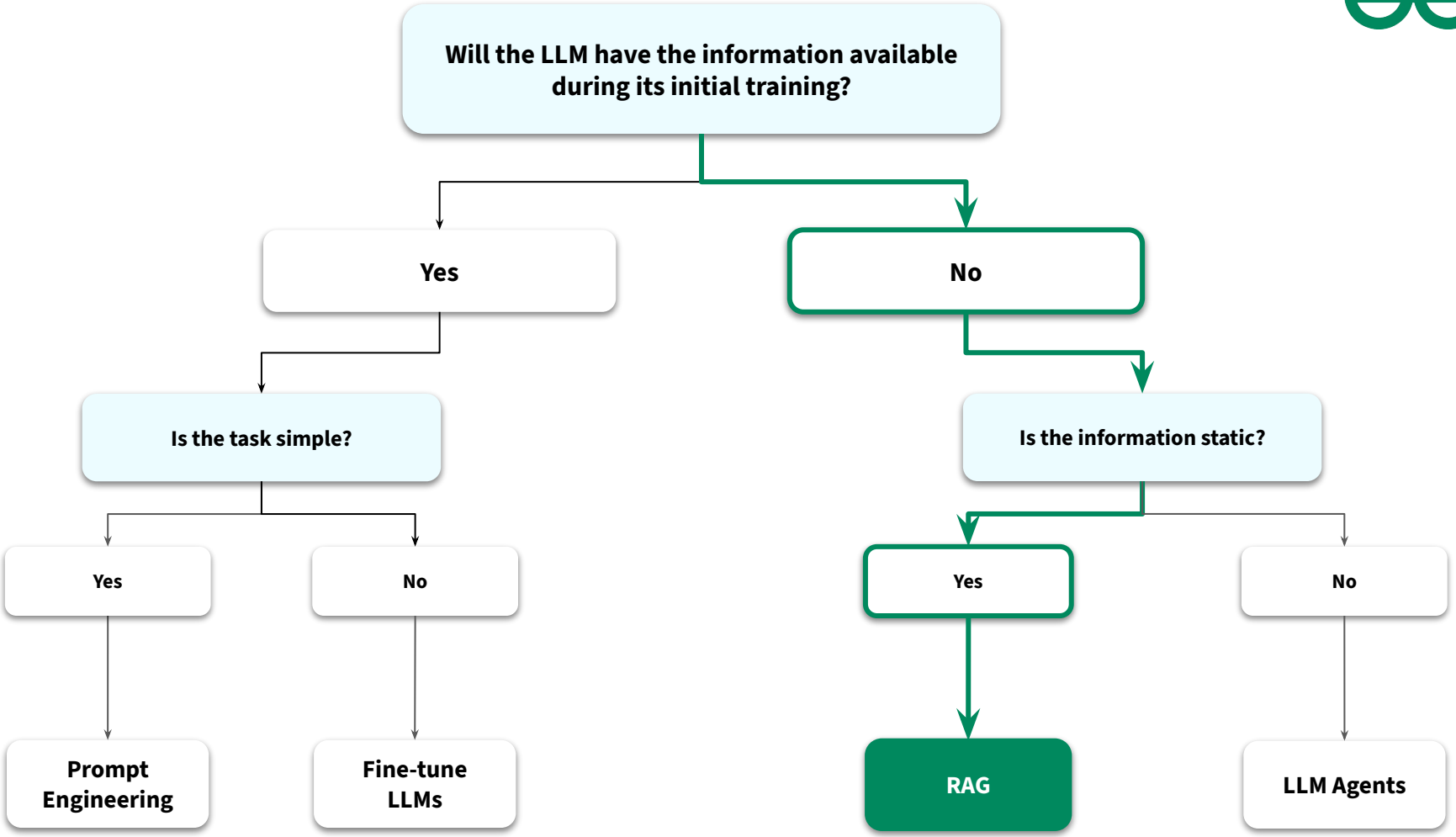
Yes
No
Prompt Engineering
Fine-tune LLMs
RAG
LLM Agents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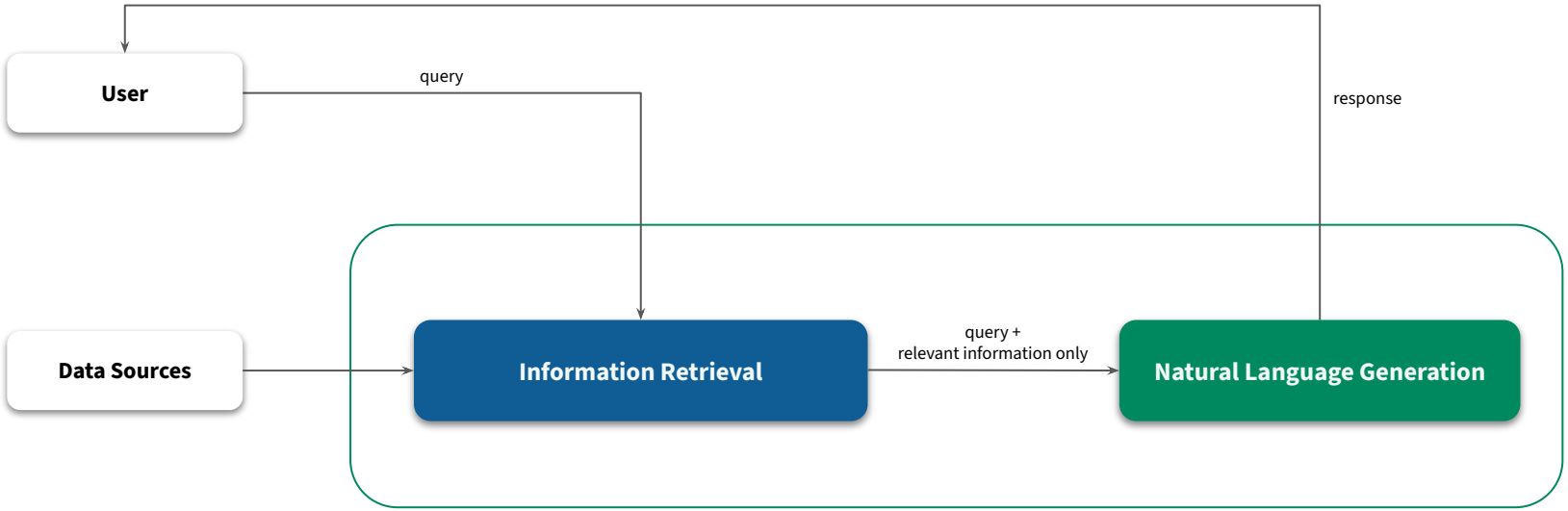
Yes
No
Prompt Engineering
Fine-tune LLMs
RAG
LLM Agents

GeeksforGeeks

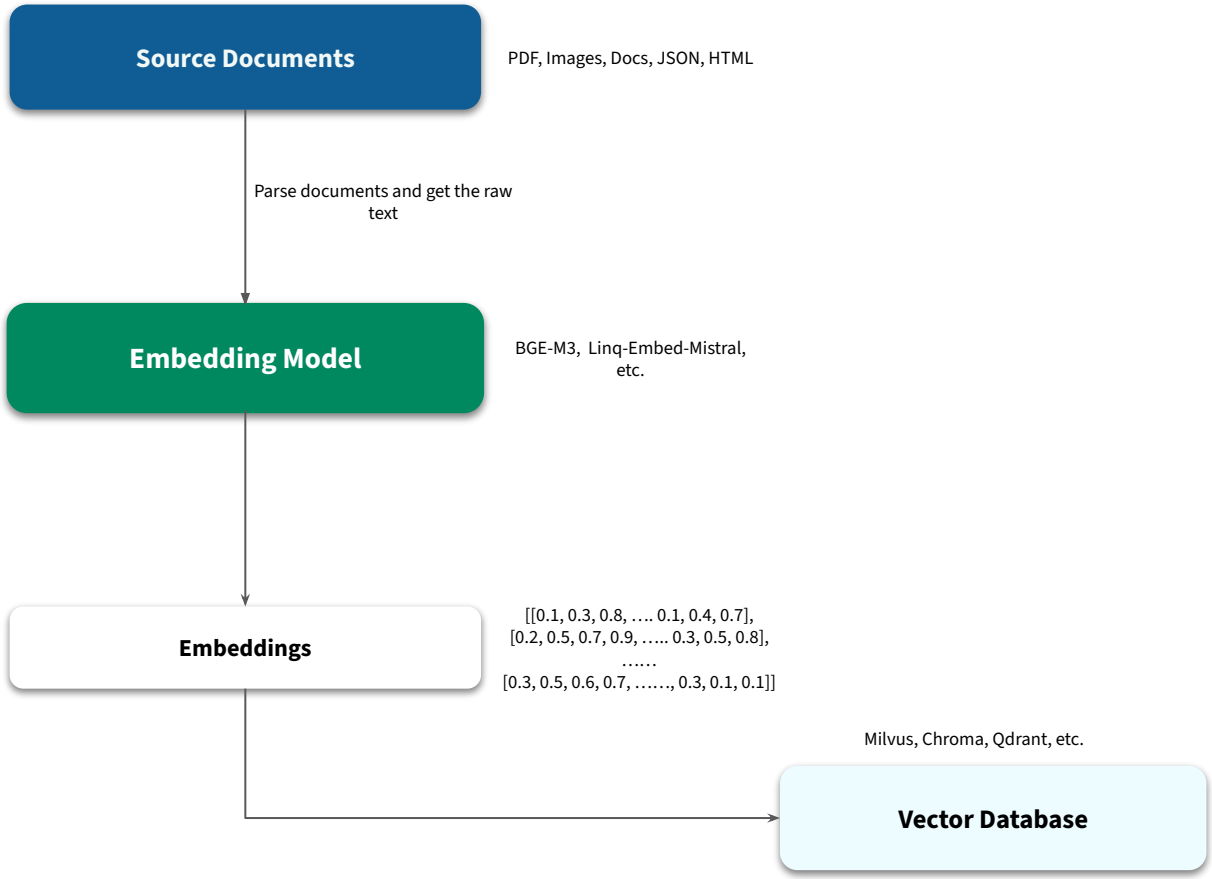
RAG Deep Dive

Retrieval-Augmented Generation

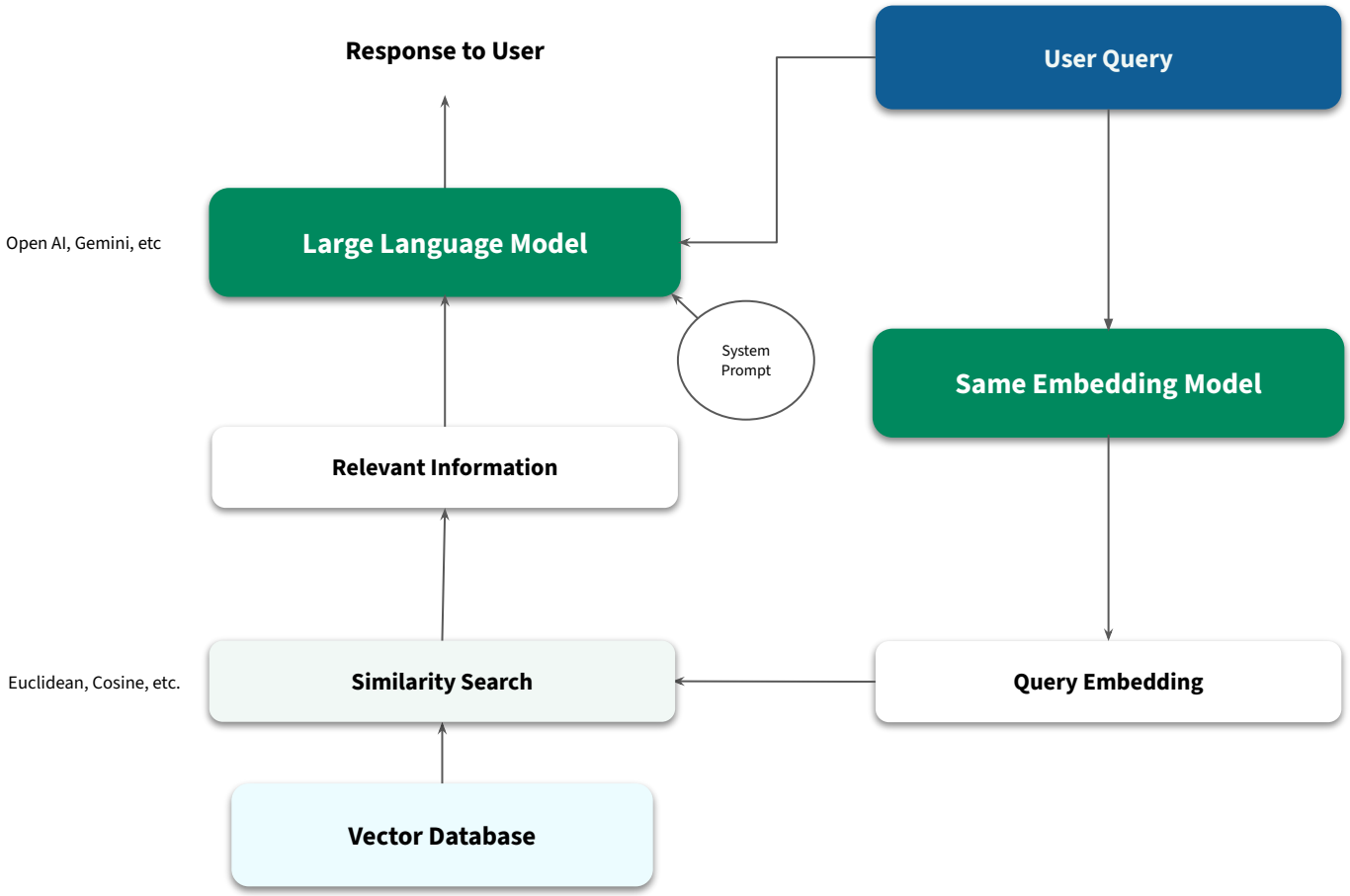
RAG


User
query
response
Data Sources
Information Retrieval
query +
relevant information only
Natural Language Generation

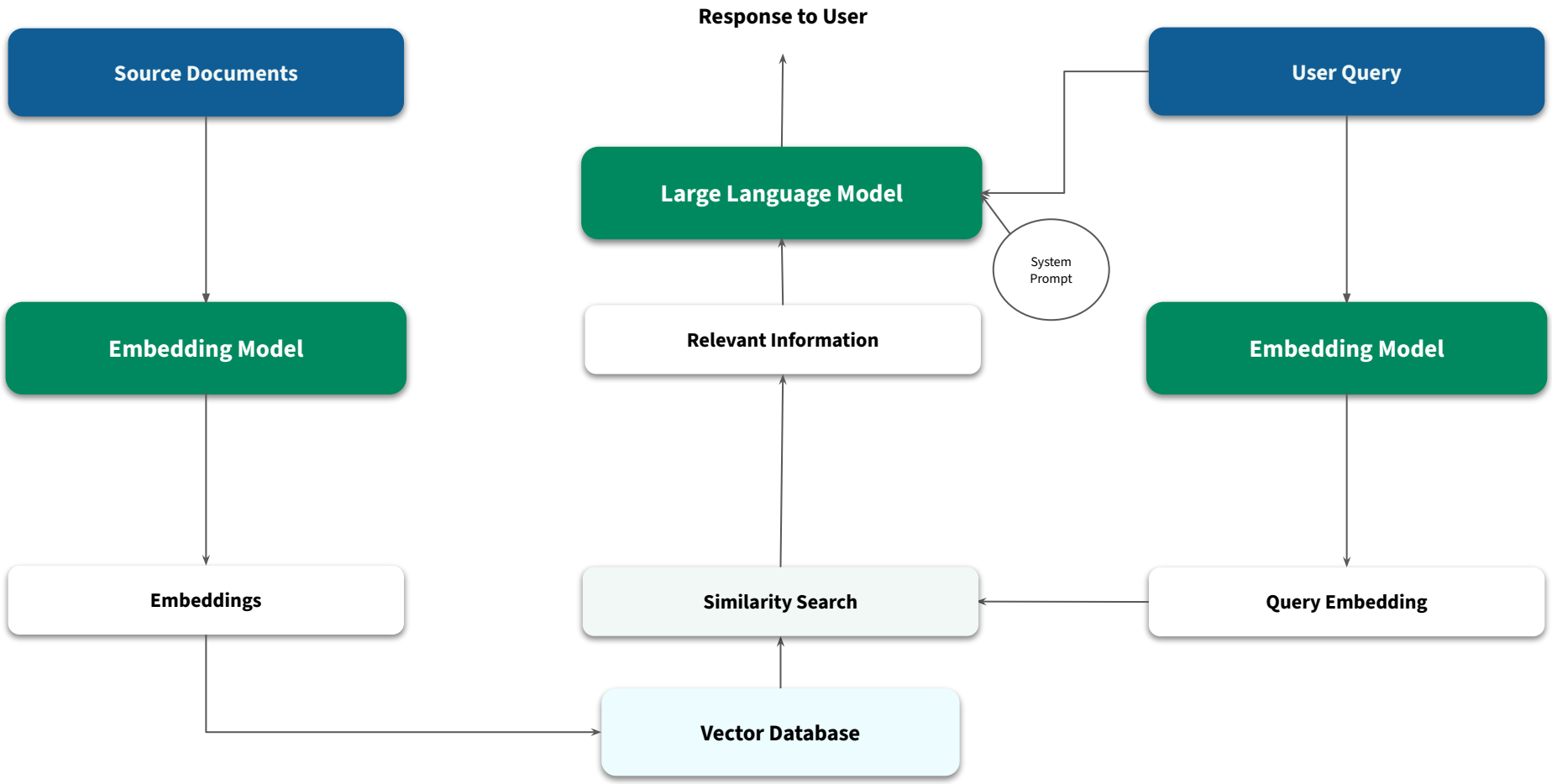
Information Retrieval: The Quest for Relevance



Natural Language Generation: The Art of Articulation



RAG Flow


Response to User
Source Documents
User Query
Large Language Model
System Prompt
Embedding Model
Relevant Information
Embedding Model
Embeddings
Similarity Search
Query Embedding
Vector Database